

# IMPERIAL

## C Project

### Group 19

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- 22/06/2026

# Emulator + Assembler

# Key Data Structures (Shared)

```
typedef enum {
    // Success
    SUCCESS,

    // Failure (stops execution/assembly)
    FAIL,

    // Emulator Specific
    INSTR_HALT,    // Halt instruction reached
    DECODE_FAIL,   // Binary failed to decode into IR
    RUNTIME_ERROR, // Invalid arguments, memory out of bound, etc.

    // Assembler Specific
    PARSE_FAIL,    // String could not be parsed into IR
    ENCODE_FAIL,   // IR could not be encoded into binary
    SYMBOL_NOT_FOUND // Forward reference missing in pass two
} ResultStatus;
```

// Internal Representation of Instructions

```
typedef struct {
    InstrGroup group;
    Opcode opcode;
    bool sf;

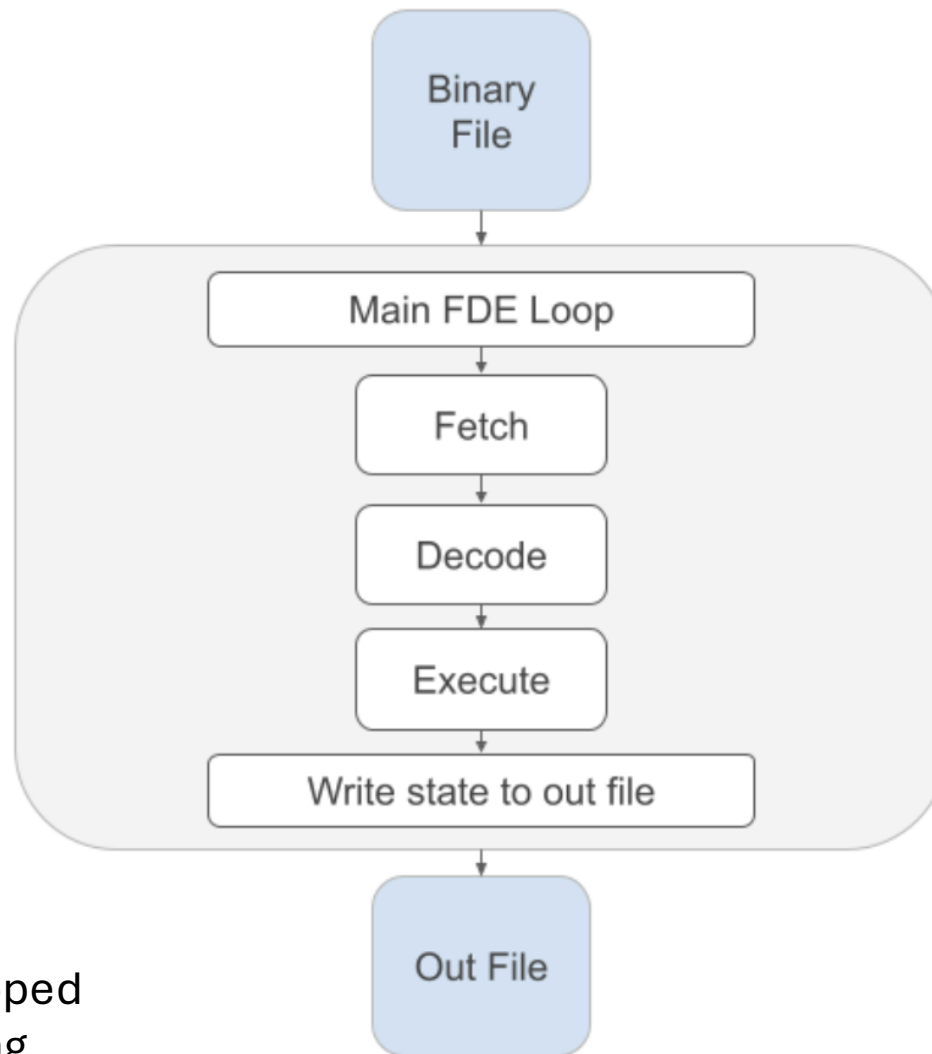
    union {
        // Data Processing (Immediate)
        struct {
            uint8_t rd;
            uint8_t rn;
            uint8_t opc;
            uint16_t imm;
            uint16_t hw;
            bool shift;
        } dp_imm;
        ...
        ...
        ...
    } Instruction_IR;
```

# Emulator Structure

Initialised registers and memory which are global variables that are updated during execution (before FDE loop begins)

Emulator

Any unknown opcodes are skipped instead of the emulator aborting



## Main FDE Loop

- Call fetch, decode and execute functions in a loop
- Checks for FAIL or INSTR\_HALT Status and stops the loop

## Fetch

- Get current instructions (4 bytes) from the input file

## Decode

- Determine main group of current instruction based on Op0
- Encode instructions as Instruction\_IR in decoders for each instruction group

## Execute

- Determine which execute function to call based on opcode
- Executes and updates program state (registers and memory)

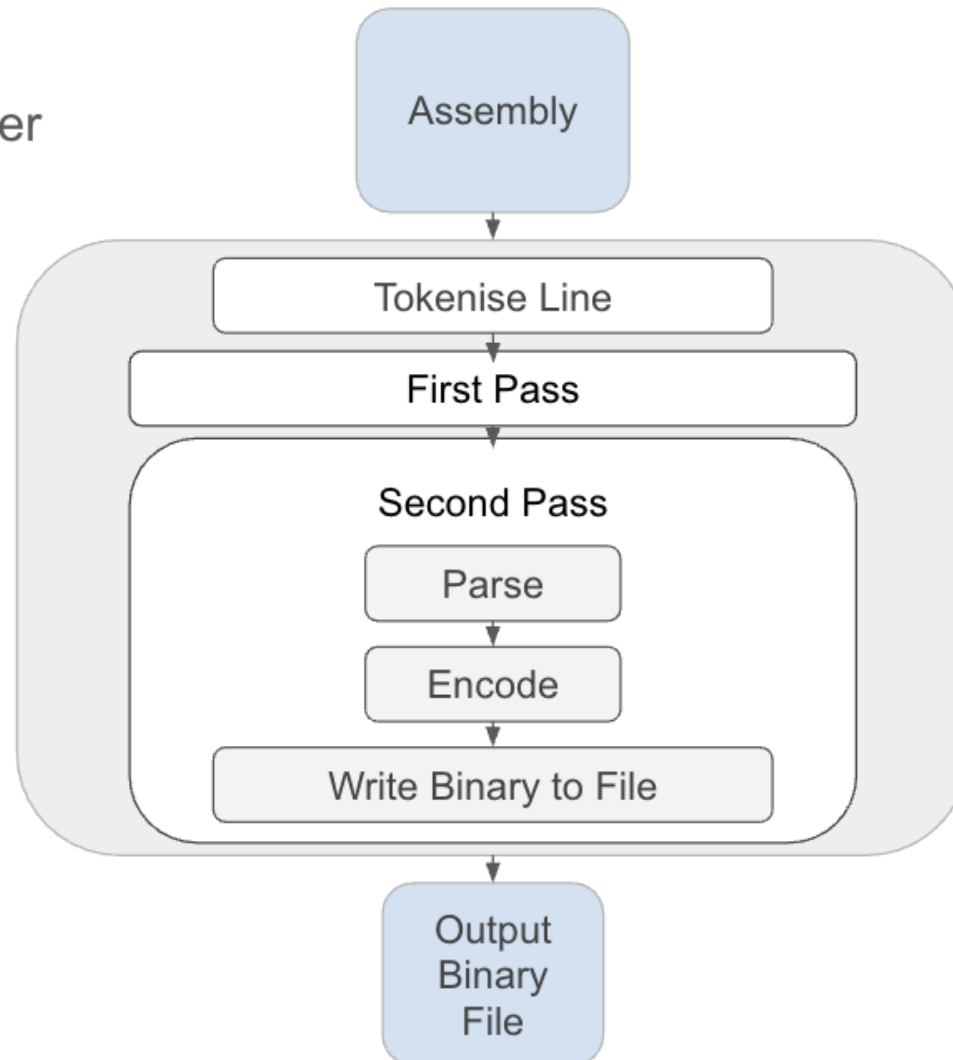
## Write state to out file

- Once out of main loop, register state and non-zero memory are outputted to a file

# Assembler Structure

## Two Pass Approach

Assembler



### Tokenise

- Split a line of assembly into label, opcode, and operands

### First Pass

- Add labels to symbol table

### Second Pass

- Determine if line is label, directive, or instruction
- Skip if it is a label or empty line
- Calls corresponding parse function if it is an instruction
- Calls encode and file writer after instruction is parsed

### Parse

- Parses instruction to `Instruction_IR` representation

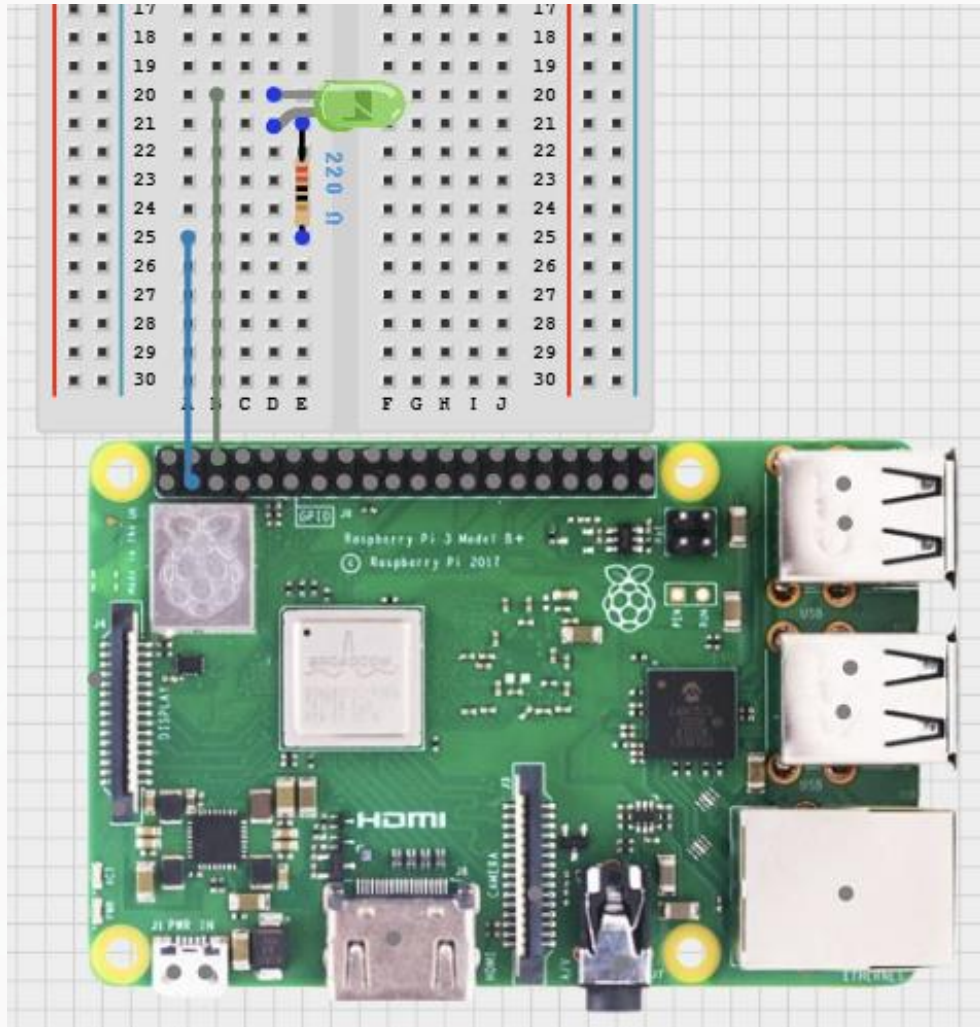
### Encode

- `Instruction_IR` is encoded into binary form

### Write Binary to File

- The binary is written to output file

# Raspberry Pi



# Lets make an LED blink! (in assembly)

- Store base address of RPI (0x3f20 0000) in w0
- Set pin 2 as output
- Store address to set pin to high/low in w2/w3 respectively
- Store value to set pin 2 to high/low in w4
- Blink at fixed delay by using decrement loop
  - Set to high
  - Set large value in w5
  - Decrement until reaches 0
  - Set low
  - Same thing as point 2
  - Repeat

# Extension



# How to English

|                    |                   |                     |                   |                    |                    |                    |                    |                     |
|--------------------|-------------------|---------------------|-------------------|--------------------|--------------------|--------------------|--------------------|---------------------|
| <b>s</b><br>sock   | <b>t</b><br>top   | <b>p</b><br>peg     | <b>n</b><br>name  | <b>m</b><br>mat    | <b>a</b><br>ant    | <b>e</b><br>ear    | <b>i</b><br>igloo  | <b>o</b><br>otter   |
| <b>g</b><br>game   | <b>d</b><br>dog   | <b>c k</b><br>clock | <b>r</b><br>rat   | <b>h</b><br>hit    | <b>u</b><br>up     | <b>ai</b><br>main  | <b>ee</b><br>tree  | <b>igh</b><br>fight |
| <b>b</b><br>bee    | <b>f</b><br>frog  | <b>l</b><br>lolly   | <b>j</b><br>jump  | <b>v</b><br>van    | <b>oa</b><br>coat  | <b>oo</b><br>look  | <b>oo</b><br>boot  | <b>ar</b><br>star   |
| <b>w</b><br>wish   | <b>x</b><br>fox   | <b>y</b><br>yell    | <b>z</b><br>zebra | <b>qu</b><br>quick | <b>or</b><br>fork  | <b>ur</b><br>burn  | <b>ow</b><br>cow   | <b>oi</b><br>coil   |
| <b>ch</b><br>chick | <b>sh</b><br>ship | <b>th</b><br>think  | <b>th</b><br>the  | <b>ng</b><br>wing  | <b>ear</b><br>fear | <b>air</b><br>hair | <b>ure</b><br>sure | <b>er</b><br>later  |

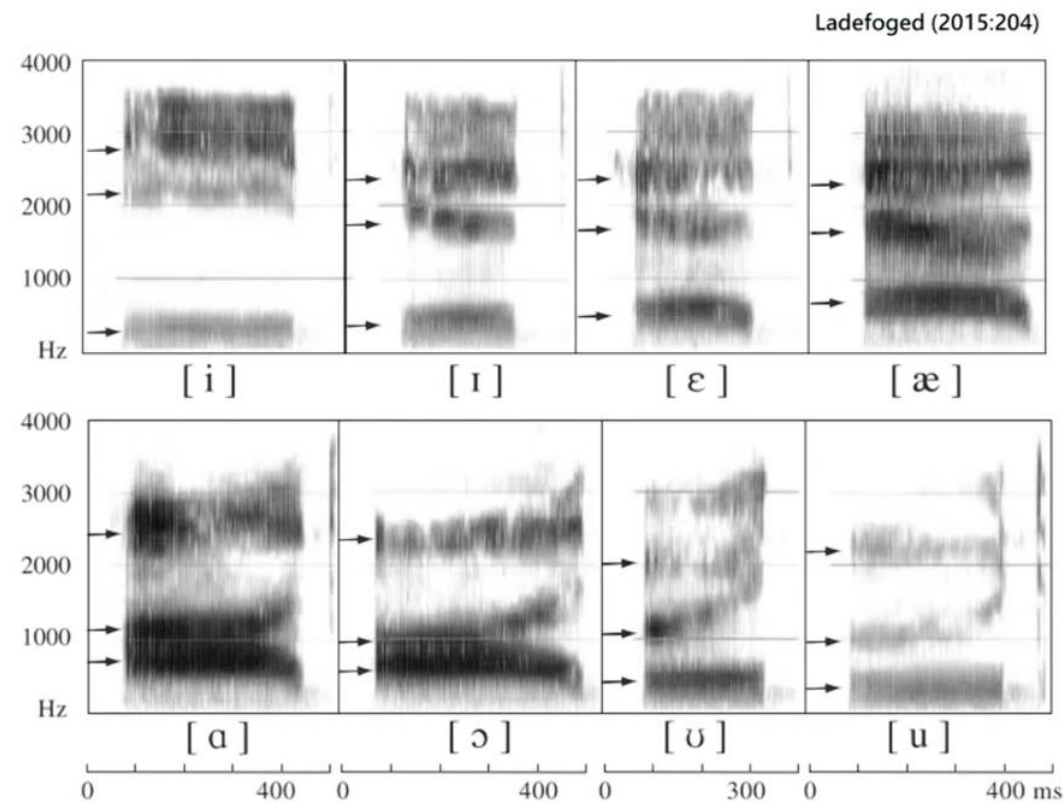
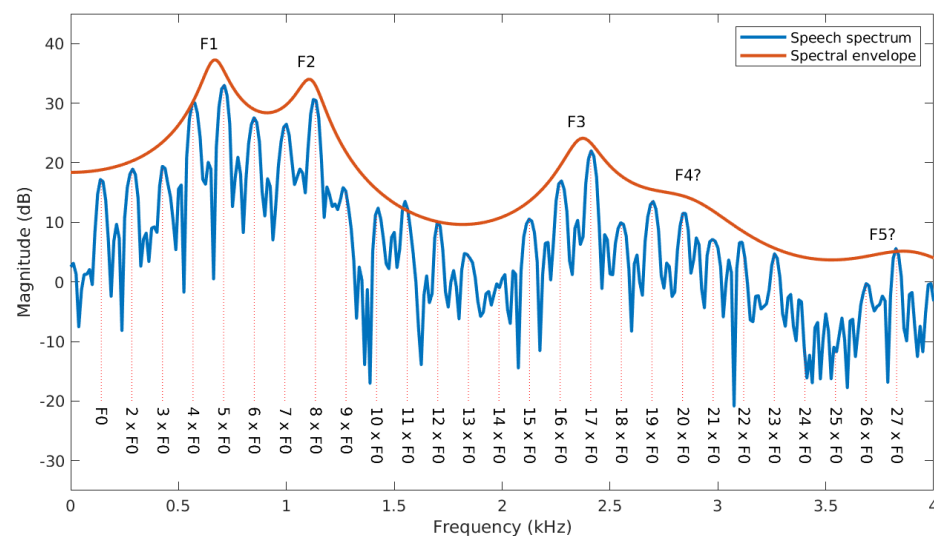
[1] What is a phoneme? [Internet]. Learning Street. Available from: <https://www.learningstreet.co.uk/articles/what-is-a-phoneme/>

| Vowel             | <i>F</i> 1 | <i>F</i> 2 | <i>F</i> 3 | <i>B</i> 1 | <i>B</i> 2 | <i>B</i> 3 |
|-------------------|------------|------------|------------|------------|------------|------------|
| [i <sup>y</sup> ] | 310        | 2020       | 2960       | 45         | 200        | 400        |
|                   | 290        | 2070       | 2960       | 60         | 200        | 400        |
| [i <sup>a</sup> ] | 400        | 1800       | 2570       | 50         | 100        | 140        |
|                   | 470        | 1600       | 2600       | 50         | 100        | 140        |
| [e <sup>y</sup> ] | 480        | 1720       | 2520       | 70         | 100        | 200        |
|                   | 330        | 2020       | 2600       | 55         | 100        | 200        |
| [e <sup>a</sup> ] | 530        | 1680       | 2500       | 60         | 90         | 200        |
|                   | 620        | 1530       | 2530       | 60         | 90         | 200        |
| [æ <sup>a</sup> ] | 620        | 1660       | 2430       | 70         | 150        | 320        |
|                   | 650        | 1490       | 2470       | 70         | 100        | 320        |
| [a]               | 700        | 1220       | 2600       | 130        | 70         | 160        |
| [ɔ <sup>a</sup> ] | 600        | 990        | 2570       | 90         | 100        | 80         |
|                   | 630        | 1040       | 2600       | 90         | 100        | 80         |
| [ʌ]               | 620        | 1220       | 2550       | 80         | 50         | 140        |
| [o <sup>ω</sup> ] | 540        | 1100       | 2300       | 80         | 70         | 70         |
|                   | 450        | 900        | 2300       | 80         | 70         | 70         |
| [u <sup>a</sup> ] | 450        | 1100       | 2350       | 80         | 100        | 80         |
|                   | 500        | 1180       | 2390       | 80         | 100        | 80         |
| [u <sup>ω</sup> ] | 350        | 1250       | 2200       | 65         | 110        | 140        |
|                   | 320        | 900        | 2200       | 65         | 110        | 140        |
| [ɐ]               | 470        | 1270       | 1540       | 100        | 60         | 110        |
|                   | 420        | 1310       | 1540       | 100        | 60         | 110        |
| [a <sup>y</sup> ] | 660        | 1200       | 2550       | 100        | 70         | 200        |
|                   | 400        | 1880       | 2500       | 70         | 100        | 200        |
| [a <sup>ω</sup> ] | 640        | 1230       | 2550       | 80         | 70         | 140        |
|                   | 420        | 940        | 2350       | 80         | 70         | 80         |
| [o <sup>y</sup> ] | 550        | 960        | 2400       | 80         | 50         | 130        |
|                   | 360        | 1820       | 2450       | 60         | 50         | 160        |

| Sonor     | <i>F</i> 1  | <i>F</i> 2  | <i>F</i> 3 | <i>B</i> 1 | <i>B</i> 2 | <i>B</i> 3 |            |            |            |            |            |    |
|-----------|-------------|-------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|----|
| [w]       | 290         | 610         | 2150       | 50         | 80         | 60         |            |            |            |            |            |    |
| [y]       | 260         | 2070        | 3020       | 40         | 250        | 500        |            |            |            |            |            |    |
| [r]       | 310         | 1060        | 1380       | 70         | 100        | 120        |            |            |            |            |            |    |
| [l]       | 310         | 1050        | 2880       | 50         | 100        | 280        |            |            |            |            |            |    |
| Fric.     | <i>F</i> 1  | <i>F</i> 2  | <i>F</i> 3 | <i>B</i> 1 | <i>B</i> 2 | <i>B</i> 3 | <i>A</i> 2 | <i>A</i> 3 | <i>A</i> 4 | <i>A</i> 5 | <i>A</i> 6 | AB |
| [f]       | 340         | 1100        | 2080       | 200        | 120        | 150        | 0          | 0          | 0          | 0          | 0          | 57 |
| [v]       | 220         | 1100        | 2080       | 60         | 90         | 120        | 0          | 0          | 0          | 0          | 0          | 57 |
| [θ]       | 320         | 1290        | 2540       | 200        | 90         | 200        | 0          | 0          | 0          |            | 28         | 48 |
| [ð]       | 270         | 1290        | 2540       | 60         | 80         | 170        | 0          | 0          | 0          | 0          | 28         | 48 |
| [s]       | 320         | 1390        | 2530       | 200        | 80         | 200        | 0          | 0          | 0          | 0          | 52         | 0  |
| [z]       | 240         | 1390        | 2530       | 70         | 60         | 180        | 0          | 0          | 0          | 0          | 52         | 0  |
| [ʃ]       | 300         | 1840        | 2750       | 200        | 100        | 300        | 0          | 57         | 48         | 48         | 46         | 0  |
| Affricate |             |             |            |            |            |            |            |            |            |            |            |    |
| [tʃ]      | 350         | 1800        | 2820       | 200        | 90         | 300        | 0          | 44         | 60         | 53         | 53         | 0  |
| [dʒ]      | 260         | 1800        | 2820       | 60         | 80         | 270        | 0          | 44         | 60         | 53         | 53         | 0  |
| Plosive   |             |             |            |            |            |            |            |            |            |            |            |    |
| [p]       | 400         | 1100        | 2150       | 300        | 150        | 220        | 0          | 0          | 0          | 0          | 0          | 63 |
| [b]       | 200         | 1100        | 2150       | 60         | 110        | 130        | 0          | 0          | 0          | 0          | 0          | 63 |
| [t]       | 400         | 1600        | 2600       | 300        | 120        | 250        | 0          | 30         | 45         | 57         | 63         | 0  |
| [d]       | 200         | 1600        | 2600       | 60         | 100        | 170        | 0          | 47         | 60         | 62         | 60         | 0  |
| [k]       | 300         | 1990        | 2850       | 250        | 160        | 330        | 0          | 53         | 43         | 45         | 45         | 0  |
| [g]       | 200         | 1990        | 2850       | 60         | 150        | 280        | 0          | 53         | 43         | 45         | 45         | 0  |
| Nasal     | <i>F</i> N1 | <i>F</i> N2 | <i>F</i> 1 | <i>F</i> 2 | <i>F</i> 3 | <i>B</i> 1 | <i>B</i> 2 | <i>B</i> 3 |            |            |            |    |
| [m]       | 270         | 450         | 480        | 1270       | 2130       | 40         | 200        | 200        |            |            |            |    |
| [n]       | 270         | 450         | 480        | 1340       | 2470       | 40         | 300        | 300        |            |            |            |    |

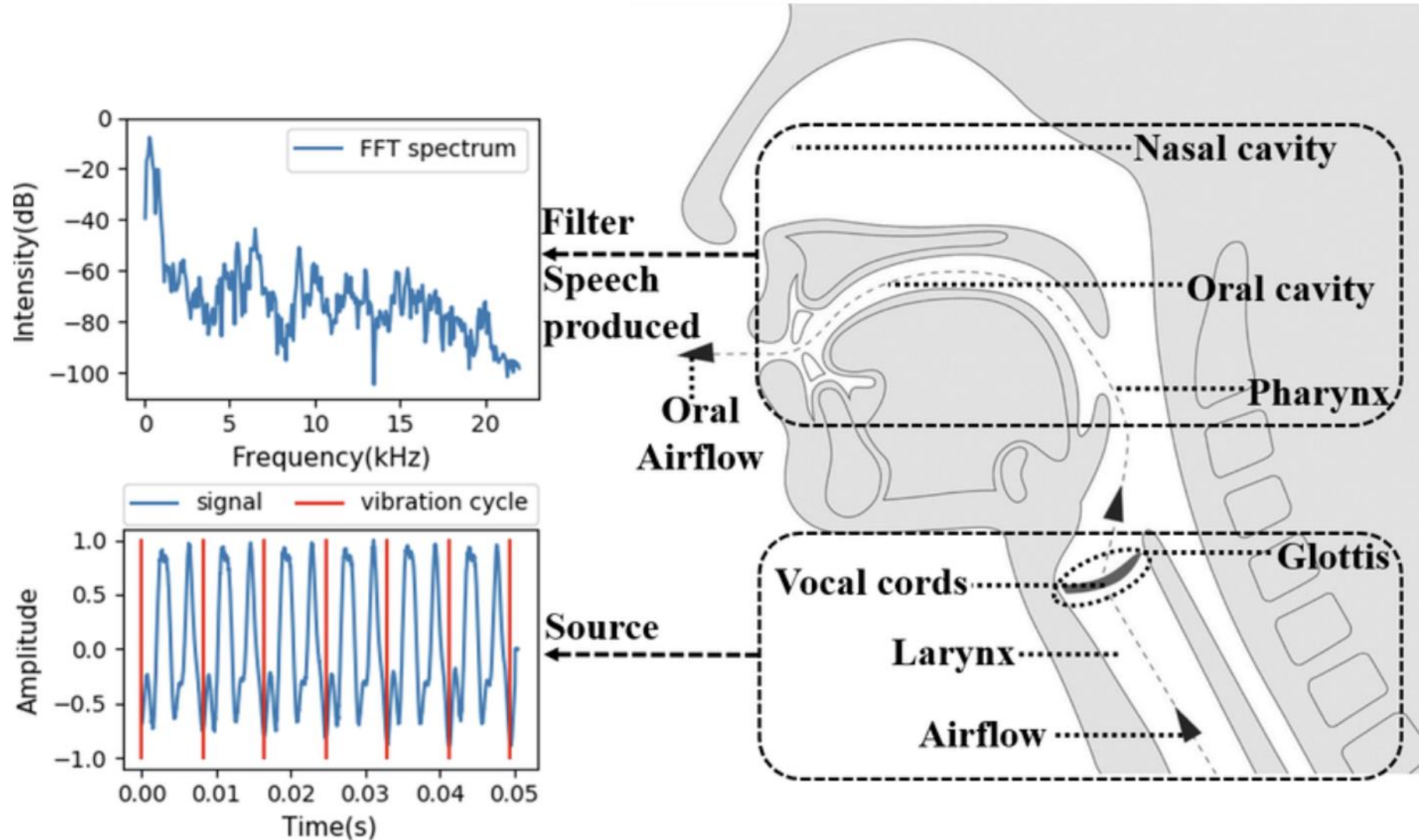
[2] Klatt DH. Software for a cascade/parallel formant synthesizer. J Acoust Soc Am.1980;67(3):971-95.

[3] Fundamental frequency (F0) — Introduction to Speech Processing [Internet]. speechprocessingbook.aalto.fi. Available from: [https://speechprocessingbook.aalto.fi/Representations/Fundamental\\_frequency\\_F0.html](https://speechprocessingbook.aalto.fi/Representations/Fundamental_frequency_F0.html)



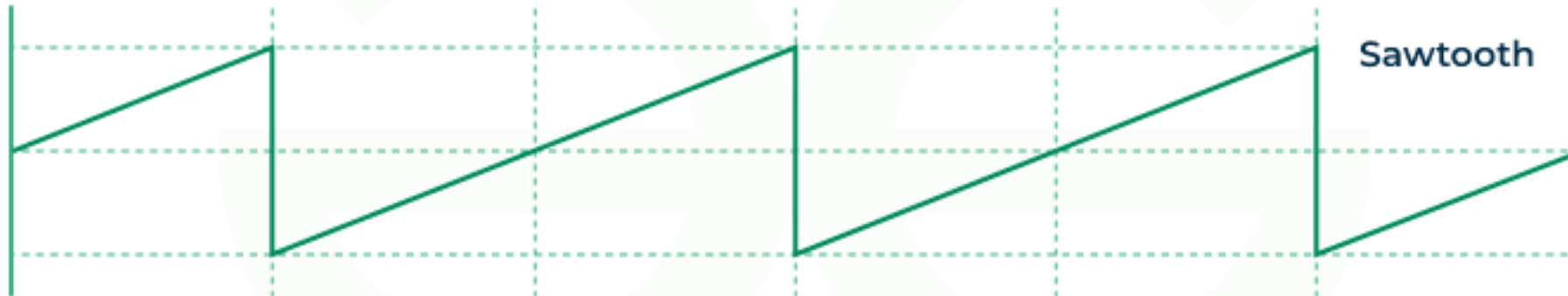
[4] EdUHK. 2.2. Formants of Vowels – Phonetics and Phonology [Internet]. Eduhk. 2025. Available from: [https://corpus.eduhk.hk/english\\_pronunciation/index.php/2-2-formants-of-vowels/](https://corpus.eduhk.hk/english_pronunciation/index.php/2-2-formants-of-vowels/)

# Glottal Source



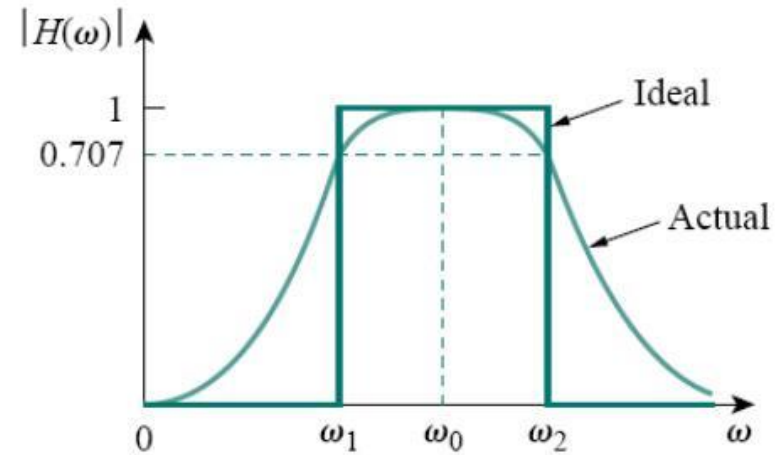
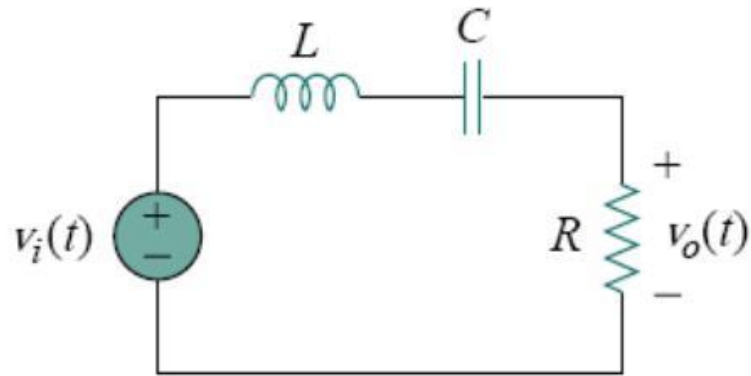
[5] Vocal organ and source-filter model [Internet]. Research Gate. Available from: [Vocal organ structure and source-filter model.](#) | [Download Scientific Diagram](#)

# Sawtooth Wave



# Bandpass Filter

A **bandpass filter** is designed to pass all frequencies within a band of frequencies,  $\omega_1 < \omega_0 < \omega_2$



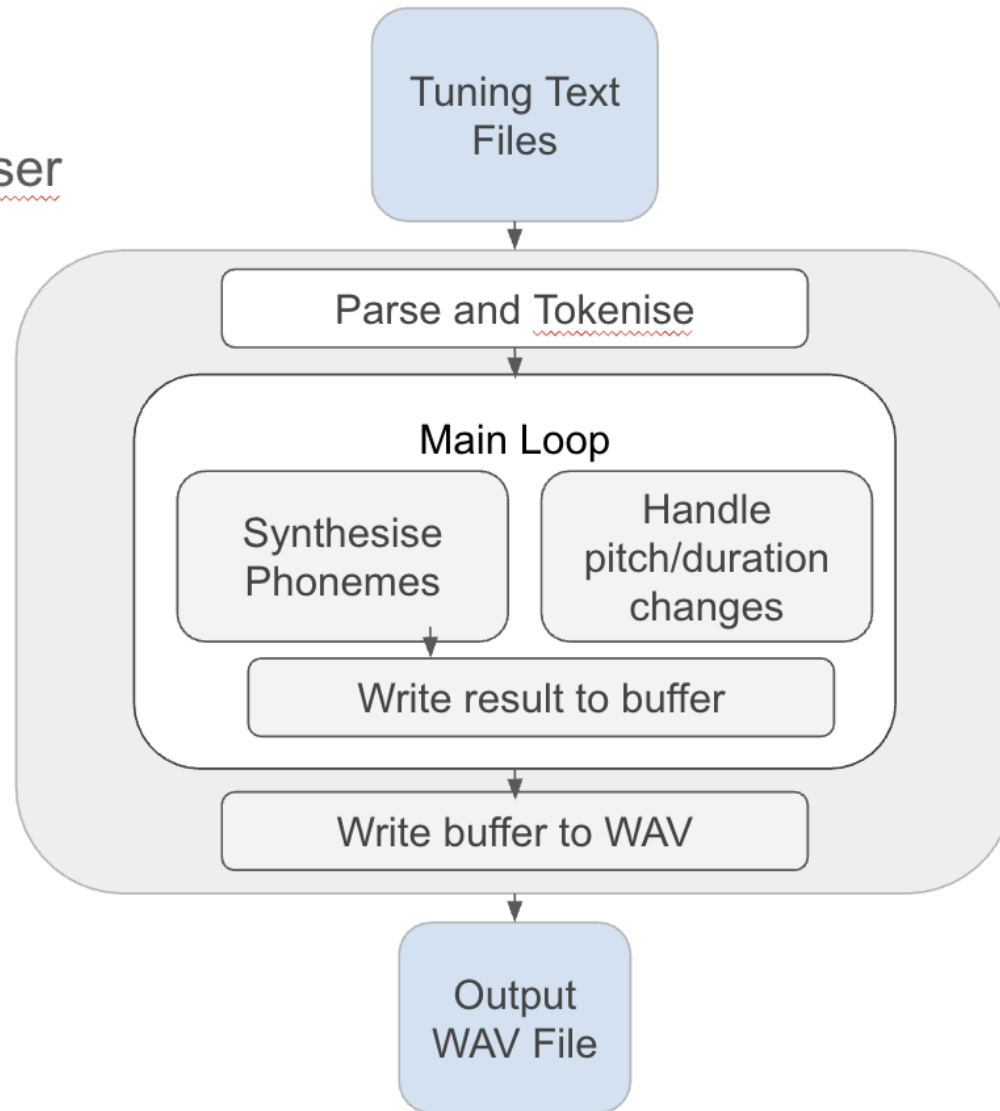
AHBMH

DEE2113 : Chap 6 - Introduction to Passive Filters



# Vocal Synth

## Vocal Synthesiser



### Parse and Tokenise

- Parse each phoneme from text file
- Parse and handles any pitch shifts, base pitch change or base duration change

### Synthesise Phonemes

- Different synthesis depending on vowel/consonant
- Runs sawtooth wave through bandpass filters
- Add all processed formants and write to buffer

### Pitch Change/Duration Changes

- Duration changes affect how many samples a phoneme takes
- Pitch change effects phase increment of oscillator

### Write buffer to WAV

- Produces resulting WAV file with sound corresponding to syntax in input



# Song 1: Daisy Bell



[8] Daisy Bell – obstinatio [Internet]. musescore. 2024. Available from: <https://musescore.com/user/4000916/scores/1104271>



[9] Scrapmanagement.com. 2026 [cited 2026 Jun 18]. Available from: <https://scrapmanagement.com/wp-content/uploads/2023/08/image-24-1024x542.png>



# Song 2: Fairytale

## Fairytale

Alexander Rybak

[10] Fairytale - F4rba [Internet]. musescore. 2024. Available from: [Fairytale – Alexander Rybak \(Violin Solo\) Sheet Music for Violin \(Solo\) | MuseScore.com](#)

♩ = 108

4

7

Years a-go, when I was youn-ger,

11

I kin-da liked a girl I knew. She was mine and we were swe-et-hearts.

15

That was then, but then it's true. I'm in love with a fai-ry-ta-le,

19

e-ven tho it hurts, \_\_\_\_\_, cause I don't care if I lose my mi-nd.

23

I'm al-ready cu - rsed.

# Song

## 3: Mockingbird

[10] Mockingbird - xZeron [Internet]. musescore.  
2024. Available from:

<https://musescore.com/user/36115541/scores/9984553>

76  
hold ya through the night. I know Mommy's not here right now and we don't know why. We feel how we feel inside. It may seem a little

79  
crazy, pretty baby but I promise Mama's gon' be alright And if you ask me to,

81  
Daddy's gonna buy you a mockingbird I'ma give you the world I'ma buy a diamond ring for you, I'ma sing for you I'll do anything

84  
for you to see you smile and if that mocking- bird don't sing and that ring don't shine I'ma break that birdie's neck I'll go back to the

87  
jeweler who sold it to ya and make him eat every carat, don't fuck with Dad!

The image displays a musical score for a piano piece in G major (one sharp). The score is presented in five systems, each with a treble and bass staff. The lyrics are written below the staves. The first system (measures 76-78) features a steady eighth-note accompaniment in the bass and a melody of eighth and sixteenth notes in the treble. The second system (measures 79-80) continues this pattern. The third system (measures 81-83) introduces a more complex treble melody with triplets and sixteenth notes. The fourth system (measures 84-86) maintains the complexity in the treble. The fifth system (measures 87-89) concludes with a final chord in the treble while the bass continues its accompaniment. The final measure (89) is highlighted with a red background.

# Group Reflection

# IMPERIAL

# Thank you

- C Project
- 22/06/2026